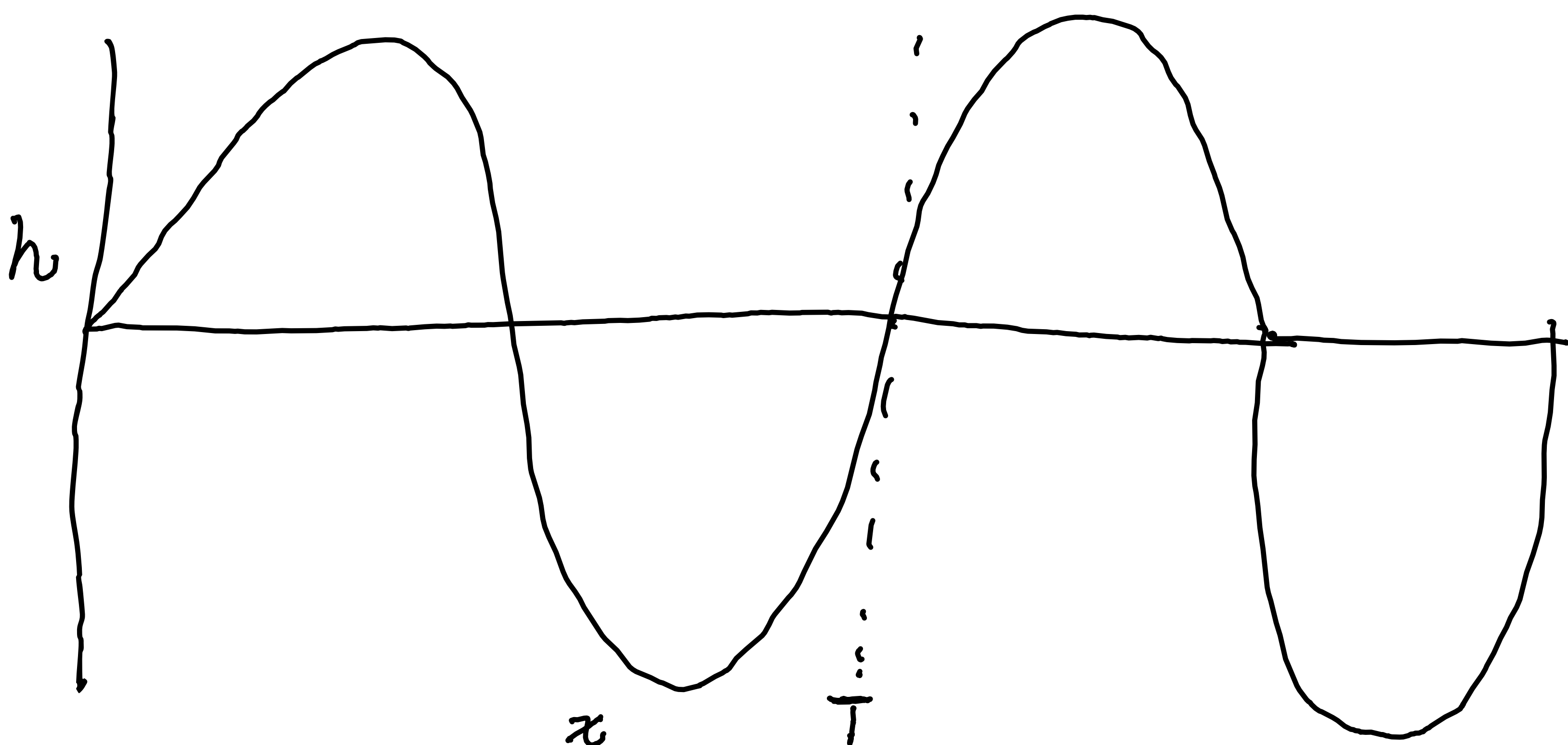


Fourier transforms primer

Imagine you have a signal like so



What is the frequency of this signal?

→ we only have a table of values, $h(x)$. We don't have a "functional form".

To find it, do a trick called the "Fourier transform"

$$\tilde{h}(q) = \frac{1}{\sqrt{L}} \int_0^L dx h(x) e^{iqx}$$

$\uparrow$ length of signal

sanity check

suppose $h(x) = \frac{1}{\sqrt{L}} \cos(\omega x)$

$$\tilde{h}(q) = \frac{1}{L} \int_0^L dx \cos(\omega x) e^{iqx}$$

for simplicity $L \rightarrow 1$ in our sanity check
doing the integral,

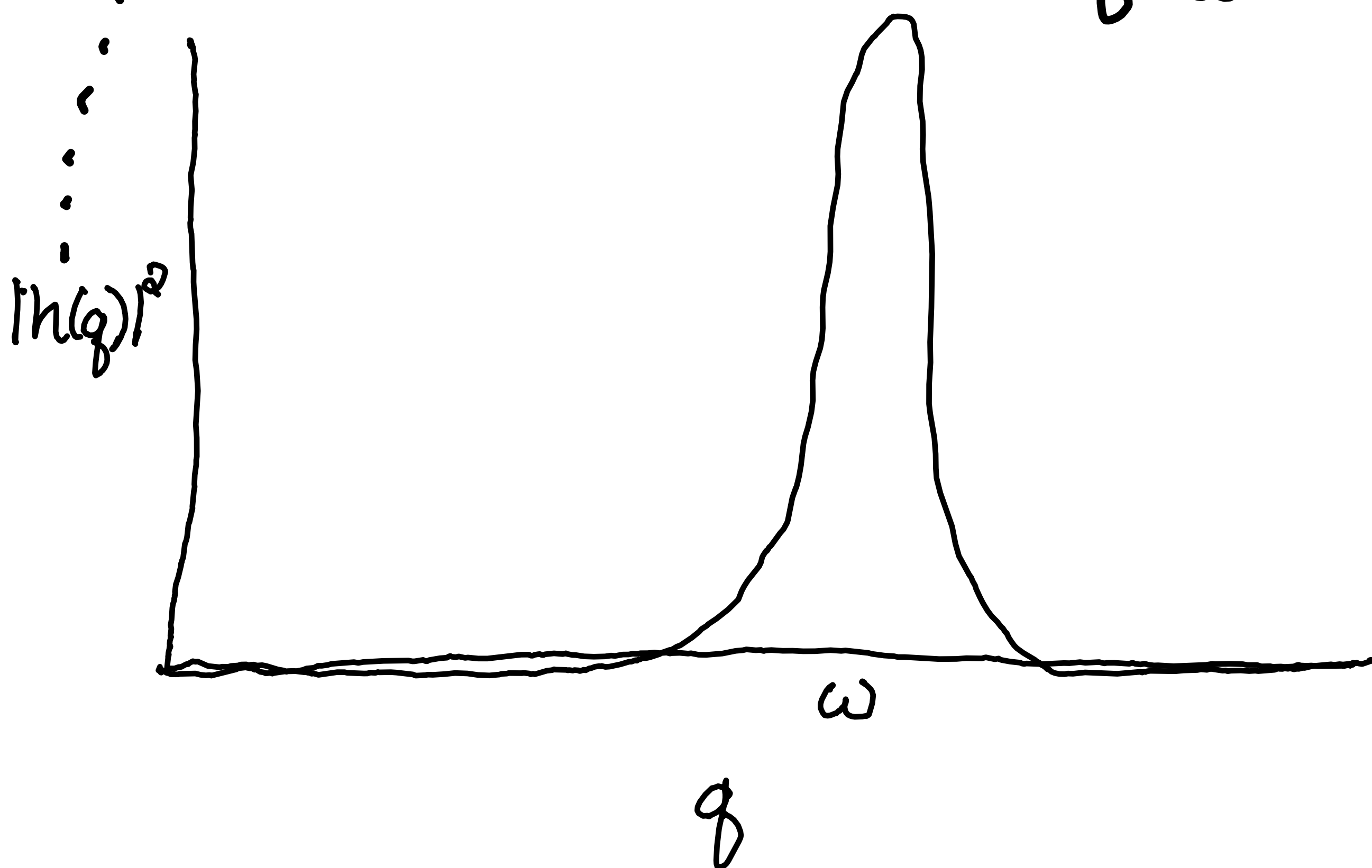
$$\tilde{h}(q) = \frac{-e^{iq}\omega \sin \omega + iq(1 - e^{iq}\cos \omega)}{q^2 - \omega^2}$$

$$\operatorname{Re}(\tilde{h}(q)) = \frac{-\omega \cos q \sin \omega + q \sin q \cos \omega}{q^2 - \omega^2}$$

$$\operatorname{Im}(\tilde{h}(q)) = \frac{-\omega \sin q \sin \omega - q \cos q \cos \omega + q}{q^2 - \omega^2}$$

$$h(q)h^*(q) = \operatorname{Re}^2 + \operatorname{Im}^2 \frac{q^2 - \omega^2}{q^2 - \omega^2} = \text{a bunch of trigonometric functions}$$

$\nwarrow$ blows up when $q = \omega$



So, if we have N numerical data points for $h(x)$, we can calculate

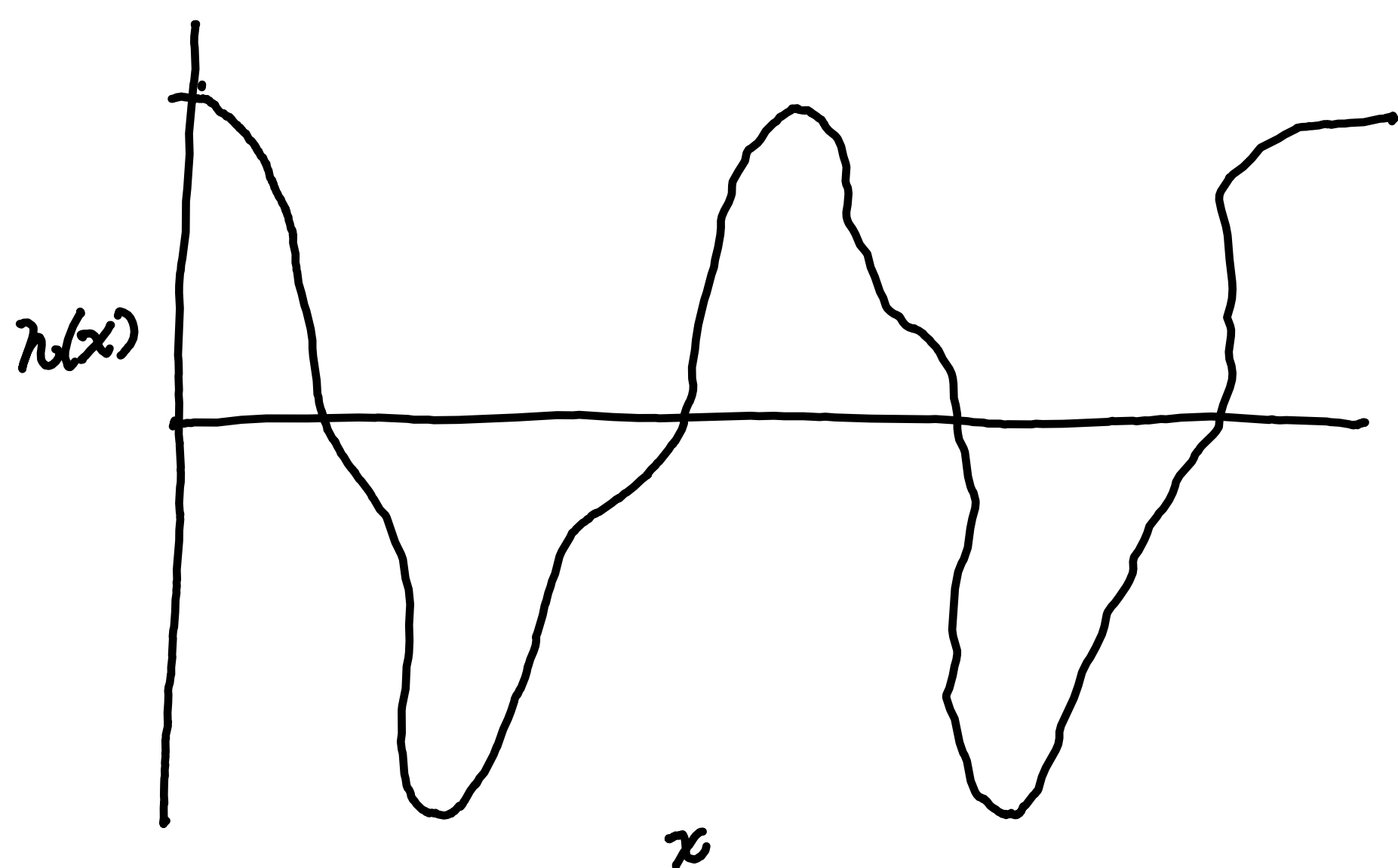
$$\lambda(q) = \frac{1}{\sqrt{N}} \sum \lambda(x) e^{iqx} \Delta x$$

Plot $|\lambda(q)|^2$, and the peak will be at ω_0'

Sanity Check 2

Suppose our numerical data represented

$$\tilde{\chi}(x) = \frac{a_1 \cos(\omega_1 x) + a_2 \cos(\omega_2 x)}{\sqrt{L}}$$

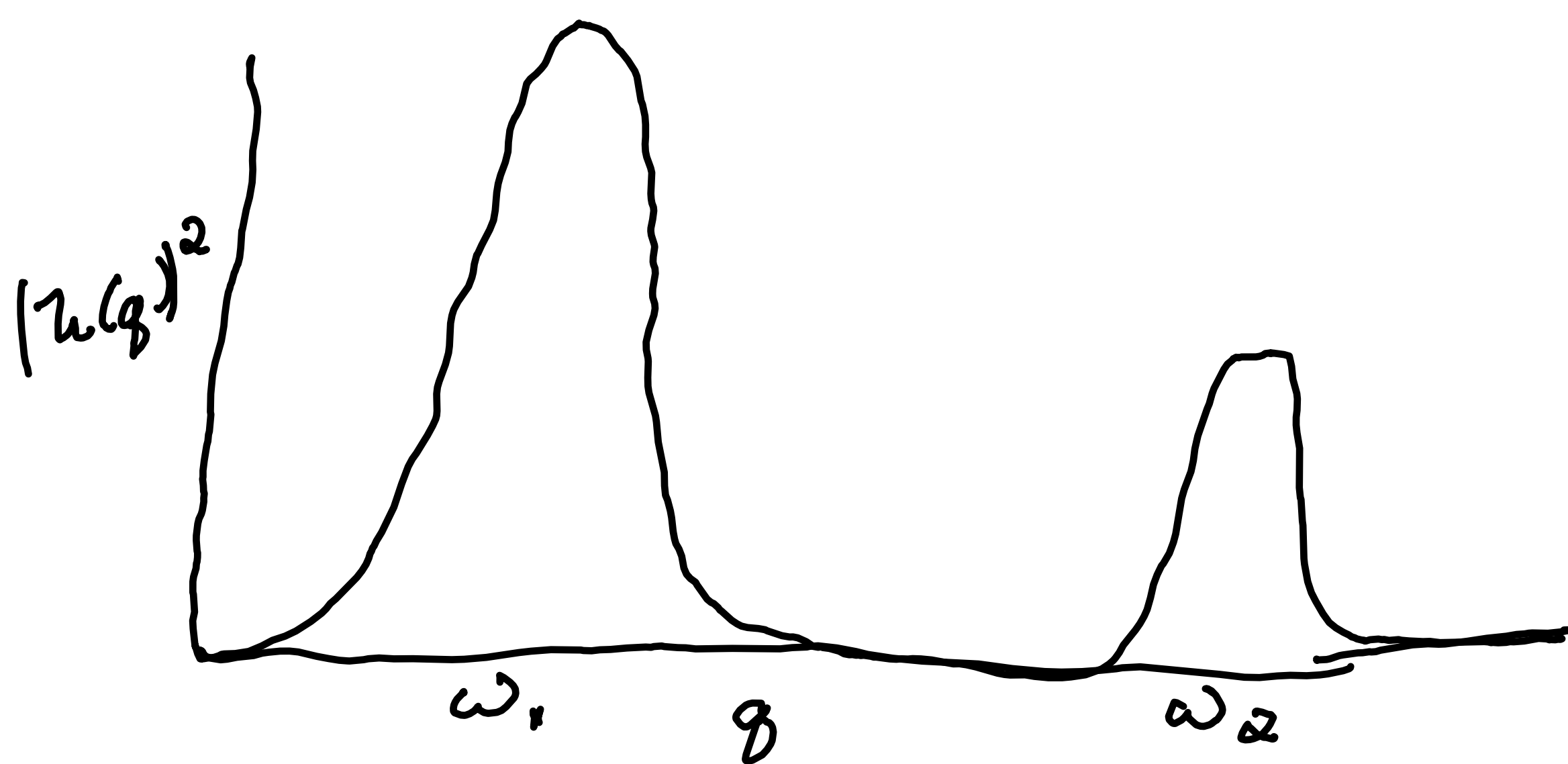


for simplicity, set $L=1$ again

$$\tilde{\chi}(q) = \int_0^1 dx \left(a_1 \cos(\omega_1 x) + a_2 \cos(\omega_2 x) \right) e^{iqx}$$

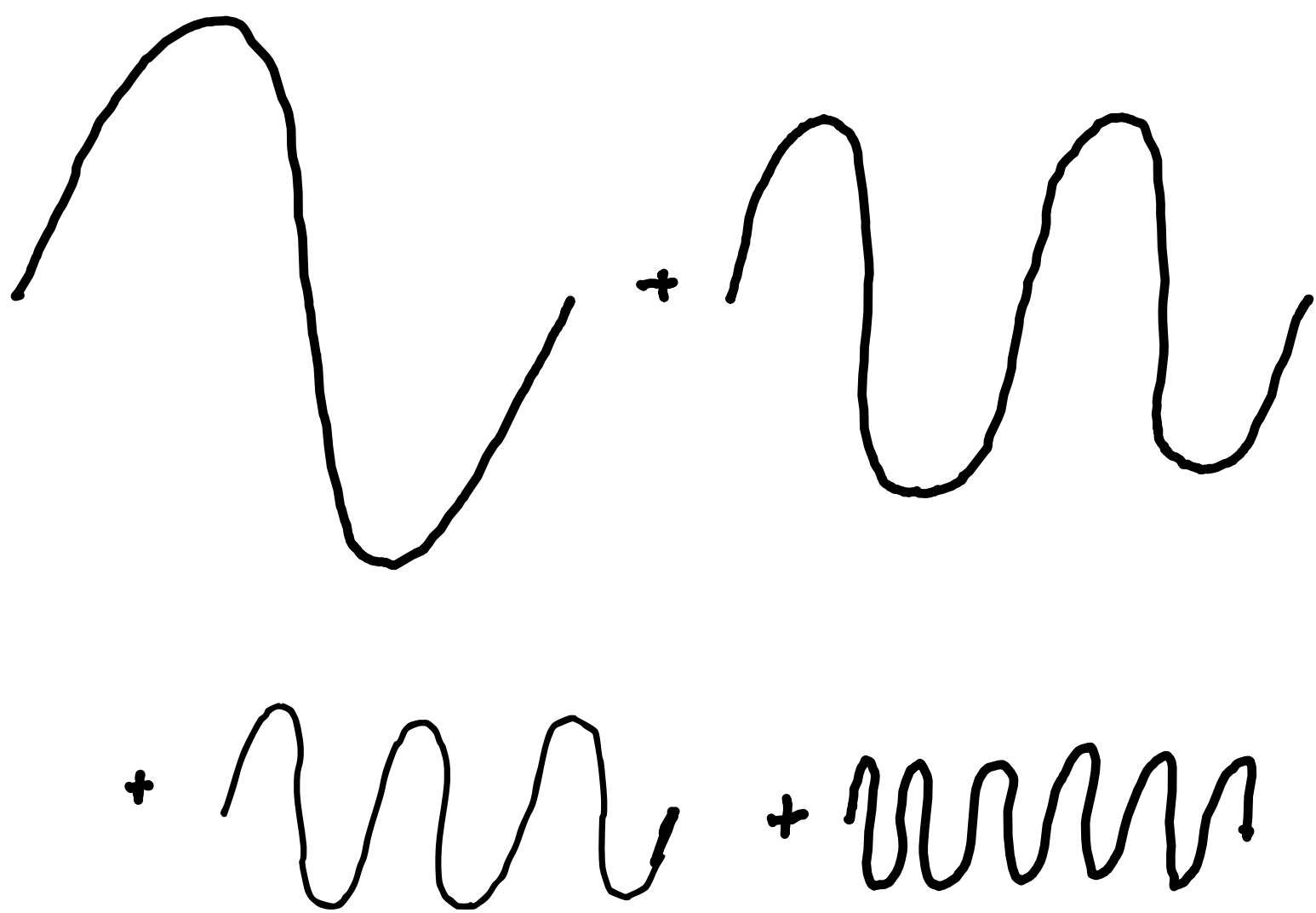
$$\tilde{\chi}(q) = a_1 \left[\frac{-e^{iq} \omega_1 \sin \omega_1 + iq(1 - e^{iq} \cos \omega_1)}{q^2 - \omega_1^2} \right] \\ - a_2 \left[\frac{e^{-iq} \omega_2 \sin \omega_2 + iq(1 - e^{-iq} \cos \omega_2)}{q^2 - \omega_2^2} \right]$$

So now $|\tilde{\chi}(q)|^2$ is going to have 2 peaks, one at ω_1 and one at ω_2 . The height of the peaks will be proportional to the peak amplitudes. If $a_1 > a_2$:

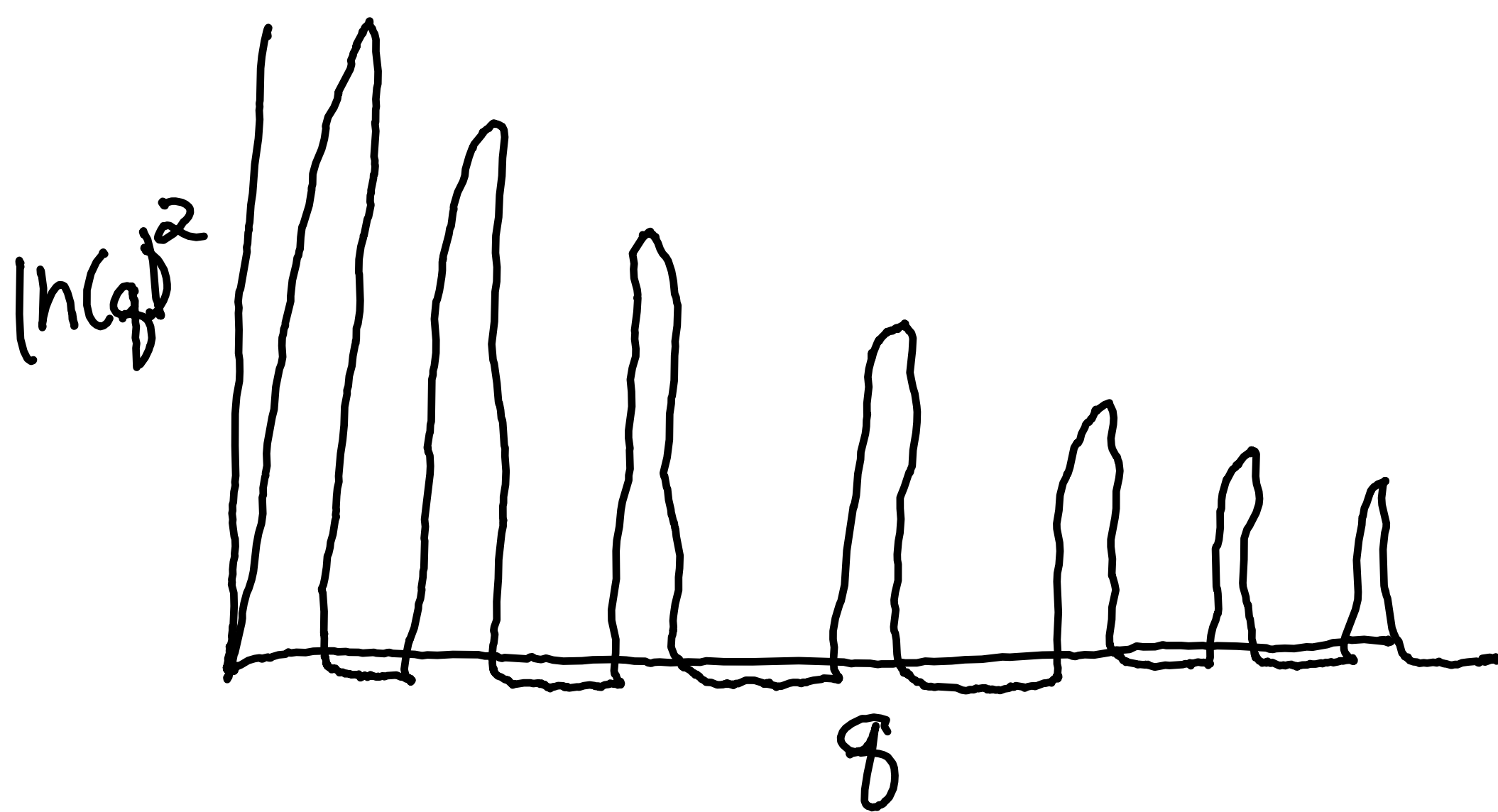


Now imagine that you had a lot of frequencies, $\omega_1, \dots, \omega_n$, and the amplitudes were a function of the frequency! i.e.

$$a_i \propto \frac{1}{\omega_i} \quad \leftarrow \text{higher frequencies have lower amplitudes}$$



Then:



or how we would normally plot it

